

● RAG Learning

Topic 5 — Light and the EM Spectrum

Edexcel GCSE (9-1) Physics (1PH0) — Paper 1

5.1 The EM Spectrum

Electromagnetic (EM) waves are **transverse** and all travel at **$3 \times 10^8 \text{ m/s}$** in a vacuum. They differ in frequency and wavelength.

Electromagnetic Spectrum — increasing frequency →

Radio	Micro-wave	Infra-red	Visible	Ultra-violet	X-rays	Gamma
$\lambda \sim \text{km}$	cm	μm	$\sim 500 \text{ nm}$	nm	0.01 nm	$< 0.01 \text{ nm}$
Uses: TV, radio	phones, cooking	heaters, thermal	vision, fibres	sterilisation	bone imaging	cancer treatment
non-ionising	non-ionising	non-ionising	non-ionising	IONISING	IONISING	IONISING

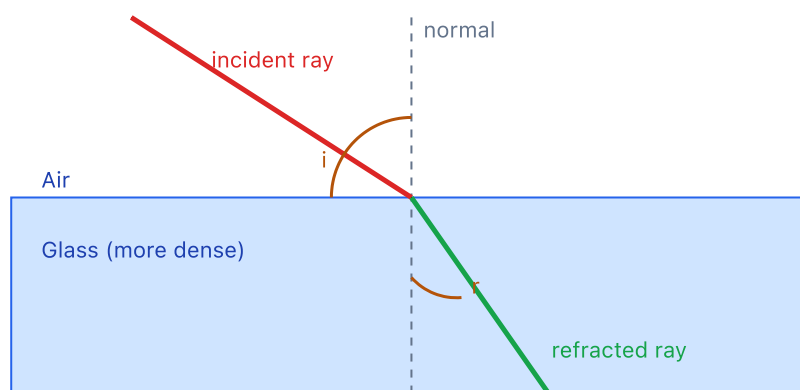
5.2 Uses and Hazards

Wave	Uses	Hazards
Radio	Broadcasting, Wi-Fi, aircraft comms	Very low risk
Microwave	Cooking, mobile phones, satellite comms	Internal heating (e.g. eyes)
Infrared	Heaters, remote controls, thermal imaging, optical fibres	Skin burns at high intensity

Visible	Vision, photography, optical fibres	Very bright light can damage eyes
Ultraviolet	Sterilisation, fluorescent lamps, sun tanning	Ionising: sunburn, cancer
X-ray	Medical imaging of bones, airport security	Ionising: mutations, cancer risk
Gamma	Sterilising equipment, killing cancer cells, medical tracers	Ionising: cell damage, mutations

5.3 Reflection and Refraction

At a boundary, light can reflect (angle of incidence = angle of reflection) or refract.



Light slows when entering glass → bends TOWARDS normal. Reverse on leaving.

5.4 Lenses

- **Convex (converging)** lenses focus parallel rays to a focal point — used in cameras, the eye, telescopes.
- **Concave (diverging)** lenses spread rays outwards; used for short-sight correction.

5.5 Colour and Filters

- A red object looks red because it *reflects* red light and absorbs others.
- A blue filter only *transmits* blue; absorbs other colours.

5.6 Total Internal Reflection

When light inside a dense medium strikes a boundary at an angle greater than the **critical angle**, all light reflects back — used in optical fibres and endoscopes.

Key Points Summary

- All EM waves travel at 3×10^8 m/s in vacuum and are transverse.
- Know uses/hazards in order: radio, micro, IR, visible, UV, X, γ .
- Higher frequency = more energy; UV, X and γ are ionising.
- Refraction = change in speed $\rightarrow$ change in direction.
- Total internal reflection above critical angle $\rightarrow$ optical fibres.